

Influenza-like illness in senior citizens:

Insights from a nursing home-based sentinel surveillance in Belgium

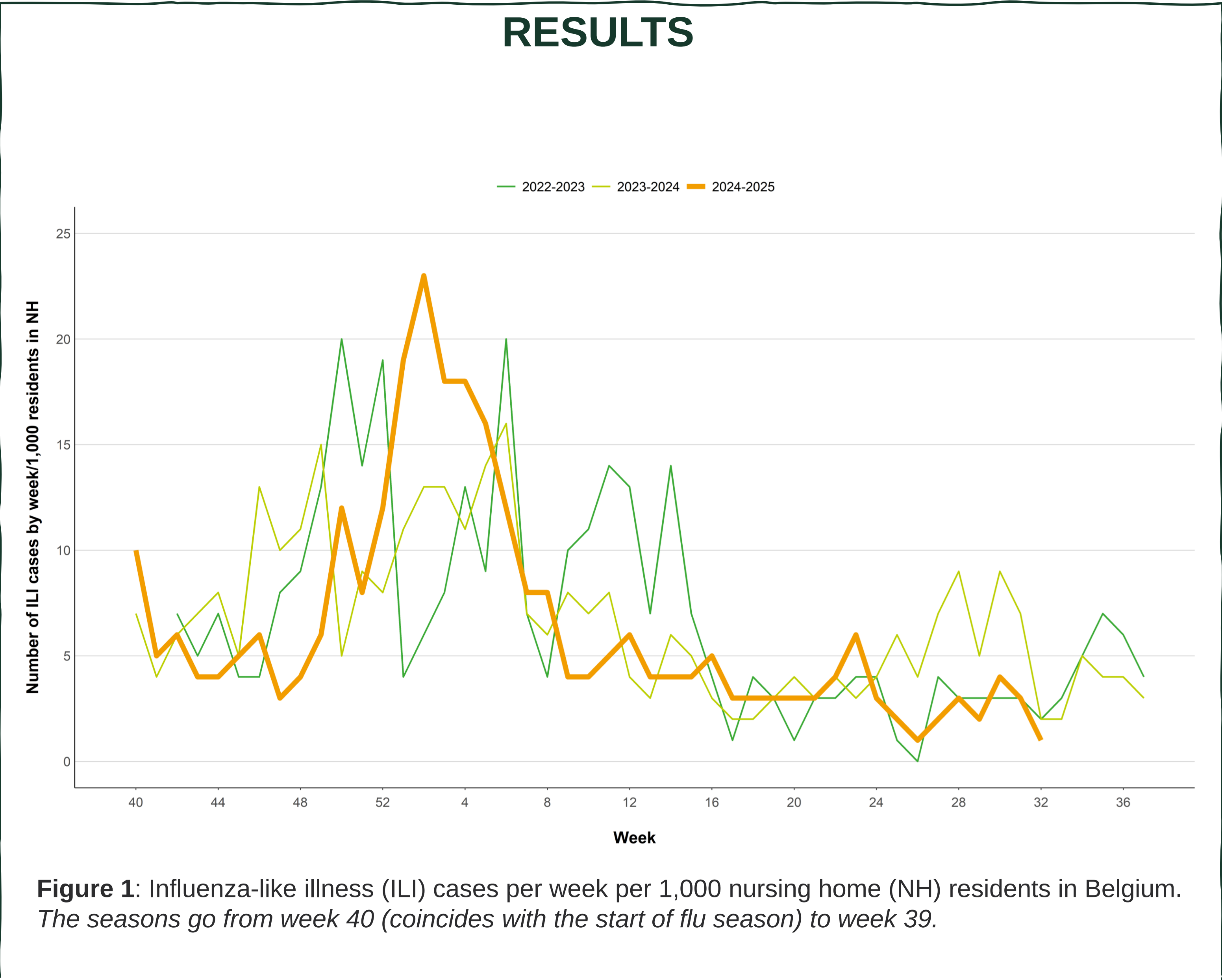
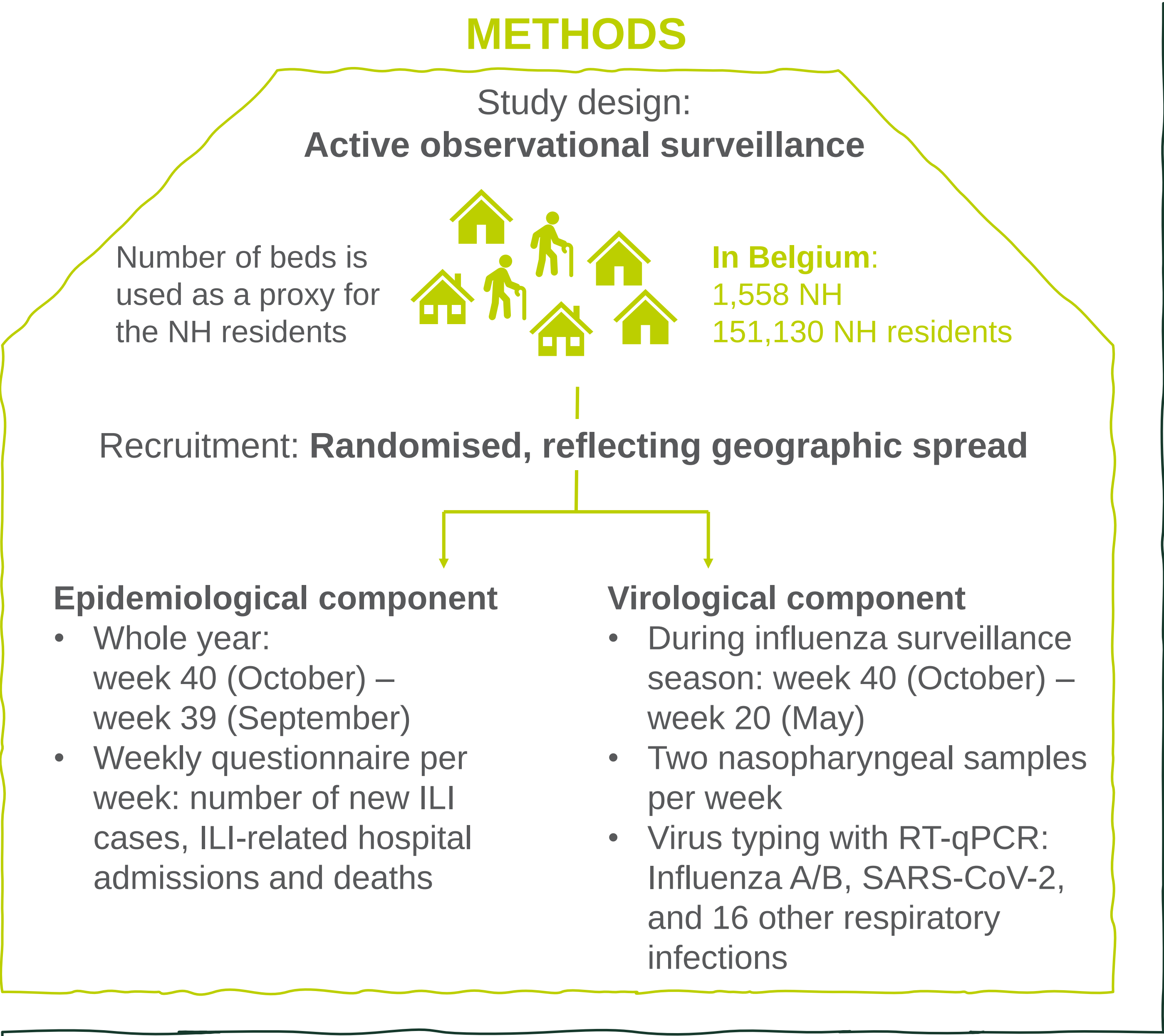
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INTRODUCTION

The COVID-19 pandemic underlined the vulnerability of the nursing home (NH) population, which accounted for about 50% of all COVID-related deaths in Belgium. In October 2022, an NH-based sentinel surveillance to monitor influenza-like illness (ILI) was set up to complement initiatives in the general population. The aim is to obtain a robust and representative network of participating NH that are geographically representative for Belgium. Results presented here are from October 2022 and onwards.



	2022-23	2023-24	2024-25*
Participation	34 NH	41 NH	68 NH
	2,943 NH residents	3,834 NH residents	6,614 NH residents
	Belgian influenza epidemic: week 49, 2022 to week 12, 2023	Belgian influenza epidemic: week 50, 2023 to week 8, 2024	Belgian influenza epidemic: week 50, 2024 to week 10, 2025
Epidemiological	Peaks in NH (Fig. 1): - Dec 2022: 20/1,000 NH residents - Feb 2023: 20/1,000 NH residents (outbreak in 1 NH)	Peaks in NH (Fig. 1): - Dec 2023: 15/1,000 NH residents - Feb 2024: 16/1,000 NH residents	Peak in NH (Fig. 1): Jan 2024: 23/1,000 NH residents
	2.8 ILI-related deaths/10,000 NH residents	1.4 ILI-related deaths/10,000 NH residents	3.4 ILI-related deaths/10,000 NH residents
	58 samples collected	98 samples collected	70 samples collected
Virological	26 negative	49 negative	37 negative
	10 SARS-CoV-2 (31%)	16 Enterovirus/Rhinovirus (33%)	12 Enterovirus/Rhinovirus (36%)
	8 Seasonal coronavirus (25%)	9 SARS-CoV-2 (18%)	9 Influenza (27%)
	7 Respiratory Syncytial Virus (22%)	8 Influenza (16%)	7 Respiratory Syncytial Virus (21%)

* Data reported here does not represent the complete season. Data up to 10/08/2025 (week 32)
NH: nursing home; ILI: influenza-like illness; SARS- CoV-2: Severe acute respiratory syndrome coronavirus 2

CONCLUSION

- The observed ILI peaks coincided with influenza epidemics each season, highlighting that this surveillance complements monitoring in the general population and supports the federal Risk Assessment Group’s mitigation proposals.
- Feedback reports enable the nursing homes and regional entities to adjust measures accordingly.
- Gradual increase of the participation results in a robust sentinel network, with precise estimation of weekly incidences.
- The virological component will be made optional due to the limited number of samples received.

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